Water Soluble Vitamins

Biological function and Disease manifestation of water and fat soluble vitamins (no structures)

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B complex

- B vitamins are a class of water-soluble vitamins that play important roles in cell metabolism
- Though these vitamins share similar names, they are chemically distinct compounds that often coexist in the same foods
- In general, dietary supplements containing all eight are referred to as a vitamin B complex.

B number	Name
Vitamin B1	Thiamine
Vitamin B2	Riboflavin
Vitamin B3	Niacin
Vitamin B5	Pantothenic acid
Vitamin B6	Pyridoxine
Vitamin B7	Biotin
Vitamin B9	Folate
Vitamin B12	Cobalamins


Vitamin B1
Thiamine
Water soluble vitamins

Vitamin B1/Thiamine

- **Aleurone layer of cereals** (food grains) is a rich source of thiamine
- Therefore whole wheat flour and unpolished hand-pound rice have better nutritive value than completely polished refined foods
- When the grains are polished, aleurone layer is usually removed
- **Yeast is** also a very good source
- Thiamine is **partially destroyed by heat**

Vitamin B1/Thiamine

Physiological Role of Thiamine

- **Pyruvate dehydrogenase:** The co-enzyme form is **thiamine pyrophosphate (TPP)** It is used in oxidative decarboxylation of alpha keto acids e.g. **pyruvate dehydrogenase** catalyzes the breakdown of pyruvate, to acetyl CoA and carbon dioxide

- **Alpha ketoglutarate dehydrogenase:** An analogous biochemical reaction that requires TPP is the oxidative decarboxylation of alpha ketoglutarate to succinyl CoA and CO2

Vitamin B1/Thiamine

Physiological Role of Thiamine

- **Transketolase:** The second group of enzymes that use TPP as co-enzyme are the transketolases, in the hexose monophosphate shunt pathway of glucose

- The main role of thiamine (TPP) is in carbohydrate metabolism. So, the **requirement of thiamine is increased along with higher intake of carbohydrates**

Vitamin B1/Thiamine

Deficiency Manifestations of Thiamine

- **Beriberi:**

- Deficiency of thiamine leads to beriberi
- It is a Singhalese word, meaning **"weakness“**
- The early symptoms are **anorexia, dyspepsia, heaviness and weakness**
- Subjects feel weak and get easily exhausted

Vitamin B1/Thiamine

Deficiency Manifestations of Thiamine

Wet beriberi:

- Here **cardiovascular manifestations** are prominent. **Edema of legs, face, trunk** and serous cavities are the main features
- **Palpitation, breathlessness and distended neck veins** are observed
- Death occurs due to **heart failure**

Vitamin B1/Thiamine

Deficiency Manifestations of Thiamine

Symptoms of **wet beriberi** include:

- increased heart rate
- severe lack of energy or constant fatigue
- shortness of breath
- waking at night due to shortness of breath
- swelling in the legs and feet

Vitamin B1/Thiamine

Deficiency Manifestations of Thiamine

Cerebral beriberi:

- **Wernicke-Korsakoff syndrome**, Clinical features are those of encephalopathy
- **ophthalmoplegia:** the paralysis or weakness of the eye muscles
- **nystagmus** a vision condition in which the eyes make repetitive, uncontrolled movements
- **cerebellar ataxia** plus **psychosis**, **loss of memory**, inability to form new memories
- **hallucinations**
- It is seen only when the nutritional status is severely affected.

Vitamin B1/Thiamine

Deficiency Manifestations of Thiamine

Dry beriberi:

- In this condition, **CNS manifestations** are the major features
- **Walking becomes difficult**
- **Peripheral neuritis** with sensory disturbance leads to complete paralysis

Vitamin B1/Thiamine

Deficiency Manifestations of Thiamine

Dry beriberi symptoms include:

- decreased muscle function, particularly in the lower legs
- tingling or loss of feeling in the feet and hands
- pain
- mental confusion
- difficulty speaking
- vomiting
- involuntary eye movement
- paralysis

Vitamin B1/Thiamine

Deficiency Manifestations of Thiamine

Polyneuritis

- It is common in **chronic alcoholics**
- Alcohol utilization needs large doses of thiamine
- **Alcohol inhibits intestinal absorption of thiamine**, leading to thiamine deficiency
- **Polyneuritis** may also be associated with pregnancy and old age
- Such thiamine deficiency in alcoholism may cause impairment of conversion of pyruvate to acetyl CoA
- This results in **increased plasma concentration of pyruvate and lactate, leading to lactic acidosis**

Vitamin B1/Thiamine

Dietary sources of thiamine

- beans and legumes, seeds
- Meat, fish, egg
- whole grains, nuts
- Dairy, yeast, nuts, legumes
- certain vegetables, such as asparagus, acorn squash, brussels sprouts, spinach, and beet greens





Vitamin B_2

Riboflavin

Water soluble vitamins

Vitamin B2/Riboflavin

Co-enzyme Activity of Riboflavin

- Riboflavin exists in **tissues tightly bound** (but not covalently) with enzymes
- Enzymes containing riboflavin are called **flavoproteins**
- The two coenzymes are **FMN (flavin mono nucleotide)** and **FAD (flavin adenine dinucleotide)**

FAD Accepts Hydrogen

- During the oxidation process, FAD accepts two hydrogen atoms from substrate. In turn, FAD is reduced to FADH2. The two nitrogen atoms of isoalloxazine nucleus accept the hydrogen atoms

Vitamin B2/Riboflavin

Co-enzyme Activity of Riboflavin

FMN-dependent Enzymes

- During the amino acid oxidation, **FMN is reduced It is reoxidized by molecular oxygen to produce hydrogen peroxide**
- In the respiratory chain**, the NADH dehydrogenase contains FMN**. The electrons are transported in the following manner

NAD+ → FMN → CoQ

Vitamin B2/Riboflavin

Co-enzyme Activity of Riboflavin

FAD-dependent Enzymes

- FADH2 when oxidized in the electron transport chain will **generate 1 ½ ATP molecules**
- Eg:- **Succinate to fumarate by succinate dehydrogenase**
- Acyl CoA to alpha-beta unsaturated acyl CoA byacyl CoA dehydrogenase
- Xanthine to uric acid by xanthine oxidase
- Pyruvate to acetyl CoA by pyruvate dehydrogenase
- Alpha ketoglutarate to succinyl CoA by alpha ketoglutarate dehydrogenase

Vitamin B2/Riboflavin

Riboflavin Deficiency

- Natural deficiency of riboflavin in man is **uncommon**, because riboflavin is **synthesized by the intestinal flora**
- Riboflavin deficiency usually **accompanies other deficiency diseases such as beriberi, pellagra and kwashiorkor**
- **Glossitis** – inflammation of tongue
- **Magenta colored tongue**
- **Cheilosis:** a painful inflammation and cracking of the corners of the mouth. It sometimes occurs on only one side of the mouth, but usually involves both sides (**Angular stomatitis**)
- **Circumcorneal vascularization (CNV)** is the in-growth of new blood vessels from the pericorneal plexus into avascular corneal tissue as a result of oxygen deprivation
- **Proliferation of the bulbar conjunctival capillaries** is the earliest sign of riboflavin deficiency

Vitamin B2/Riboflavin

Dietary Sources of Riboflavin

- Rich sources are **liver, dried yeast, egg and whole milk**
- Good sources are **fish, whole cereals, legumes and green leafy vegetables**

Daily Requirement

- Riboflavin is concerned mainly in the metabolism of carbohydrates and requirement is **related to calorie intake**
- Adults on sedentary work require about **1.5 mg per day.**
- During pregnancy, lactation and old age, **additional 0.2 to 0.4 mg /day are required.**





Vitamin B_3

Niacin

Water soluble vitamins

Vitamin B3/Niacin

- Niacin and Nicotinic acid are synonyms. It is also called as **Pellagra preventing factor** of Goldberger.
- Niacin is **pyridine-3-carboxylic acid**. **Niacinamide is the acid amide**

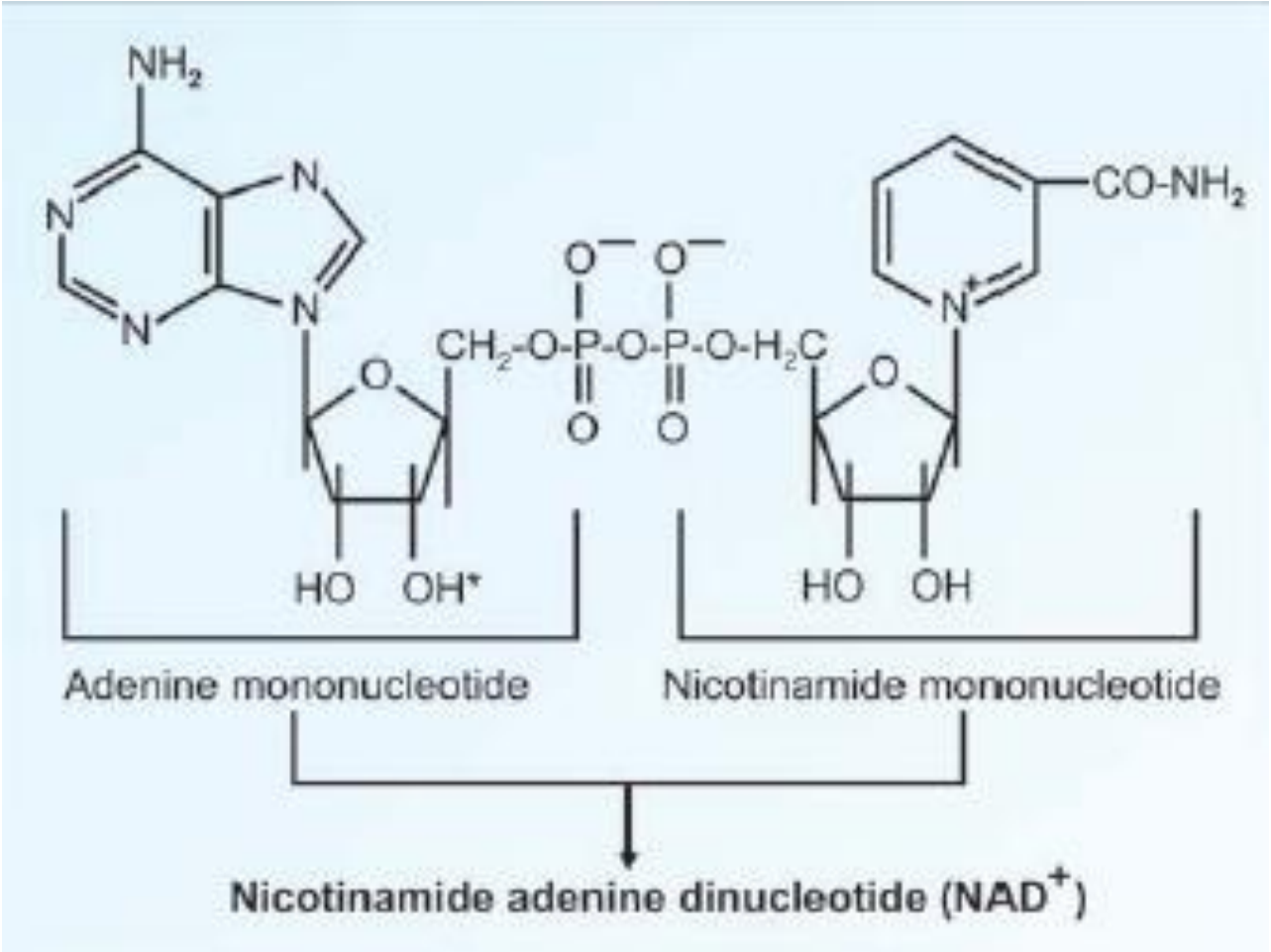
Co-enzyme Forms of Niacin

- Niacin is converted to its co-enzyme forms, viz. **Nicotinamide adenine dinucleotide (NAD+)** and **Nicotinamide adenine dinucleotide phosphate (NADP+)**
- The niacin is attached to a ribose phosphate to form a mononucleotide. It is then attached to AMP, to form the dinucleotide

Vitamin B3/Niacin

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Vitamin B3/Niacin

NAD+ dependent enzymes

- Lactate dehydrogenase (lactate → pyruvate)
- Glyceraldehyde-3-phosphate dehydrogenase (glyceraldehyde-3-phosphate → 1, 3-bisphosphoglycerate)
- Pyruvate dehydrogenase (pyruvate → acetyl CoA)
- Alpha ketoglutarate dehydrogenase (alpha ketoglutarate → succinyl CoA
- Beta hydroxy acyl CoA dehydrogenase (beta hydroxy acyl CoA → beta keto acyl CoA)
- Glutamate dehydrogenase (Glutamate → alpha keto glutarate

Vitamin B3/Niacin

Pellagra

- Deficiency of niacin leads to the clinical condition called **pellagra**
- Pellagra is an Italian word, meaning **"rough skin“**
- Pellagra is caused by the **deficiency of Tryptophan as well as Niacin**
- Pellagra is **seen more in women**; this may be because tryptophan metabolism is **inhibited by estrogen metabolites**
- The symptoms of pellagra are:

Vitamin B3/Niacin

Dermatitis, Diarrhea, Dementia

- **Dermatitis:** In early stages, bright red erythema occurs, especially in the **feet, ankles and face**
- Increased pigmentation around the neck is known as **Casal's necklace**
- The dermatitis is **precipitated by exposure to sunlight**







Vitamin B3/Niacin

Diarrhea:

- The diarrhea may be mild or severe with blood and mucus
- This may lead to **weight loss**
- **Nausea and vomiting** may also be present

Dementia:

- It is frequently seen in chronic cases
- **Delerium** is common in acute pellagra.
- **Irritability, inability** to concentrate and **poor memory** are more common in mild cases.
- **Ataxia** and **spasticity** are also seen.

Vitamin B3/Niacin

Niacin is Synthesized from Tryptophan

- Quinolinate phosphoribosyl transferase is the rate limiting enzyme in the conversion of niacin to NAD+
- About 60 mg of tryptophan is equivalent to 1 mg of niacin

Vitamin B3/Niacin

Causes for Niacin Deficiency

- **Dietary deficiency of Tryptophan:** Pellagra is seen among people whose staple diet is maize (South and Central America)
- In maize, niacin is present; but it is in a bound form, and is unavailable
- Pellagra is also seen when staple diet is sorghum (jowar or guinea corn) as in Central and Western India





Vitamin B3/Niacin

Causes for Niacin Deficiency

- **Deficient synthesis:** Kynureninase, an important enzyme in the pathway of tryptophan, is pyridoxal phosphate dependent
- So conversion of tryptophan to niacin is not possible in pyridoxal deficiency
- **Isoniazid (INH):** It is an anti-tuberculous drug, which inhibits **pyridoxal phosphate formation**. Hence there is block in conversion of tryptophan to NAD+.

Vitamin B3/Niacin

Causes for Niacin Deficiency

- **Hartnup disease:** Tryptophan absorption from intestine is defective in this congenital disease
- Moreover, tryptophan is excreted in urine in large quantities. This leads to lack of tryptophan and consequently deficiency of nicotinamide
- **Carcinoid syndrome:** The tumor utilizes major portion of available tryptophan for synthesis of serotonin; so tryptophan is unavailable

Vitamin B3/Niacin

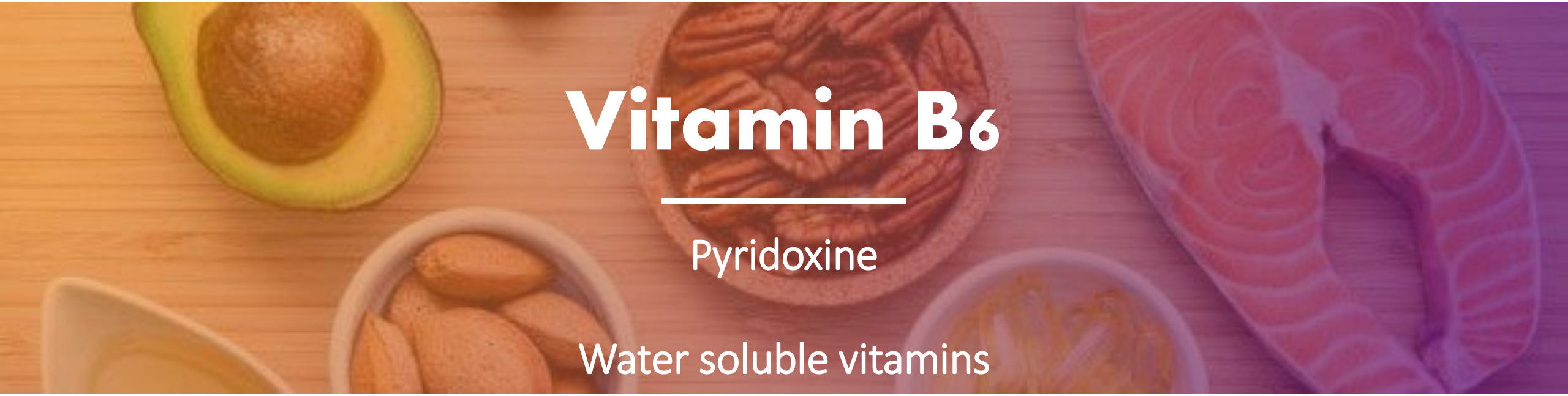
Dietary Sources of Niacin

- The richest natural sources of niacin are **dried yeast, rice polishing, liver, peanut, whole cereals, legumes, meat and fish**
- About **half** of the requirement is met by the **conversion of tryptophan to niacin**
- About 60 mg of tryptophan will yield 1 mg of niacin



Recommended Daily Allowance (RDA)

Normal requirement is 20 mg/day. During lactation, additional 5 mg are required



Vitamin B6

Pyridoxine

Water soluble vitamins

Vitamin B6/Pyridoxine

Co-enzyme Form

- Vitamin B6 is the term applied to a family of 3 related pyridine derivatives; **pyridoxine** (alcohol), **pyridoxal** (aldehyde) and **pyridoxamine**
- Active form of pyridoxine is **pyridoxal phosphate (PLP)**

Vitamin B6/Pyridoxine

Functions of Pyridoxal Phosphate

- The pyridoxal phosphate (PLP) acts as co-enzyme for many reactions in amino acid metabolism

Transamination

- These reactions are catalyzed by **amino transferases** (transaminases) which employ PLP as the co-enzyme. For example:

Alanine + Alpha keto glutarate ⟶ **Pyruvate + Glutamic acid**

(Enzyme Alanine transaminase)

Vitamin B6/Pyridoxine

Decarboxylation

- All decarboxylation reactions of amino acids require PLP as co-enzyme
- Active form of pyridoxine is **pyridoxal phosphate (PLP)**

Sulfur Containing Amino Acids

- PLP plays an important role in methionine and cysteine metabolism

Heme Synthesis

- ALA synthase is a PLP dependent enzyme. This is the rate limiting step in heme biosynthesis

Vitamin B6/Pyridoxine

Production of Niacin

- Pyridoxal phosphate is required for the synthesis of niacin from tryptophan (one vitamin is necessary for synthesis of another vitamin)

Glycogenolysis

- Phosphorylase enzyme (glycogen to glucose-1- phosphate) requires PLP. In fact, more than 70% of total PLP content of the body is in muscles, where it is a part of the phosphorylase enzyme

Vitamin B6/Pyridoxine

Deficiency Manifestations of Pyridoxine

- Neurological Manifestations In vitamin B6 deficiency, PLP dependent enzymes function poorly
- So, **serotonin, epinephrine, noradrenalin and gamma amino butyric acid (GABA) are not produced properly**
- **Neurological symptoms** are therefore quite common in B6 deficiency
- In children, B6 deficiency leads to **convulsions** due to decreased formation of GABA
- PLP is involved in the synthesis of sphingolipids; so B6 deficiency leads to **demyelination of nerves** and **consequent peripheral neuritis**. This is reversible with high doses of B6.

Vitamin B6/Pyridoxine

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Vitamin B6/Pyridoxine

Dermatological Manifestations

- Deficiency of B6 will also **affect tryptophan metabolism**
- Since niacin is produced from tryptophan, B6 deficiency in turn leads to niacin deficiency which is manifested as pellagra

Vitamin B6/Pyridoxine

Dermatological Manifestations

- Hematological Manifestations
- In adults hypochromic microcytic anemia may occur due to the inhibition of heme biosynthesis
- Impaired antibody formation is also reported

Vitamin B6/Pyridoxine

Dietary Sources of Vitamin B6

- Rich sources are yeast, rice polishing, wheat germs, cereals, legumes (pulses), oil seeds, egg, milk, meat, fish and green leafy vegetables



Requirement of B6

- recommended that adults need 1 to 2 mg/day. During pregnancy and lactation, the requirement is increased to 2.5 mg/day.

Vitamin B5/Pantothenic Acid

- The Greek word “pantos” means everywhere
- As the name suggests, it is widely distributed in nature
- Pantothenic acid contains beta alanine and D-pantoic acid in amide linkage

Vitamin B5/Pantothenic Acid

Co-enzyme Activity of Pantothenic Acid

- The beta mercaptoethanol amine (NH2-CH2- CH2-SH) contains one thiol or sulfhydryl (-SH) group
- It is the active site where acyl groups are carried
- Therefore the co-enzyme A is sometimes abbreviated as CoA-SH to denote this active site
- The thio ester bond in acyl-CoA is a high energy bond
- These acyl groups are transferred to other acceptors, for example:

Acetyl CoA + Choline → Acetyl choline + CoA (enzyme is acetyl choline synthase)

Vitamin B5/Pantothenic Acid

Co-enzyme Activity of Pantothenic Acid

- Acyl groups are also accepted by the CoA molecule during the metabolism of other substrates, for example:

Pyruvate + CoA + NAD^+ → Acetyl-CoA + CO2 + NADH

(Enzyme is pyruvate dehydrogenase)

Vitamin B5/Pantothenic Acid

Co-enzyme Activity of Pantothenic Acid

- The important CoA derivatives are:

a) Acetyl CoA

b) Succinyl CoA

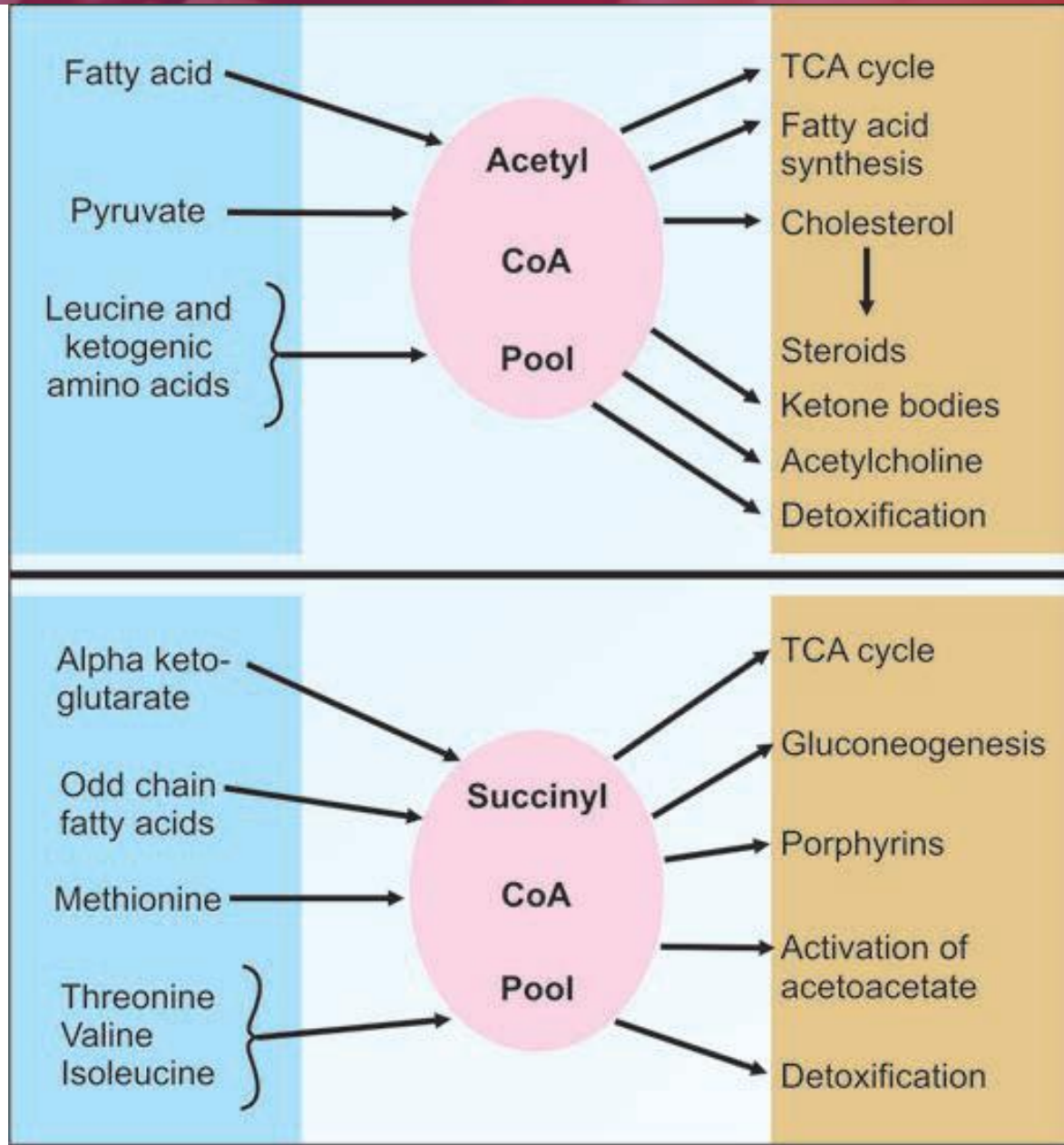
c) HMG CoA

d) Acyl CoA.

Vitamin B5/Pantothenic Acid

Co-enzyme Activity of Pantothenic Acid

- Co-enzyme A is an important component of fatty acid synthase complex
- The ACP (acyl carrier protein) also contains pantothenic acid



Vitamin B5/Pantothenic Acid

Deficiency of Pantothenic Acid

- paresthesia (burning, lightning pain) in lower extremities, staggering gait due to impaired coordination and sleep disturbances
- These deficiency manifestations are rare in human beings
- The syndrome is seen during famine, in prison camps, in chronic alcoholics and in some renal dialysis patients
- In experimental animals, deficiency has resulted in anemia (due to reduced heme synthesis from succinyl CoA), reduced steroidogenesis (due to lack of acetyl CoA), dermatitis, fatty liver and adrenal necrosis.

Vitamin B5/Pantothenic Acid

Sources of Pantothenic Acid

- It is widely distributed in plants and animals
- Moreover, it is synthesized by the normal bacterial flora in intestines
- Therefore, deficiency is very rare
- Yeast, liver and eggs are good sources



Requirement of Pantothenic Acid

- RDA is assumed to be about 10 mg/day



Vitamin B7

Biotin

Water soluble vitamins

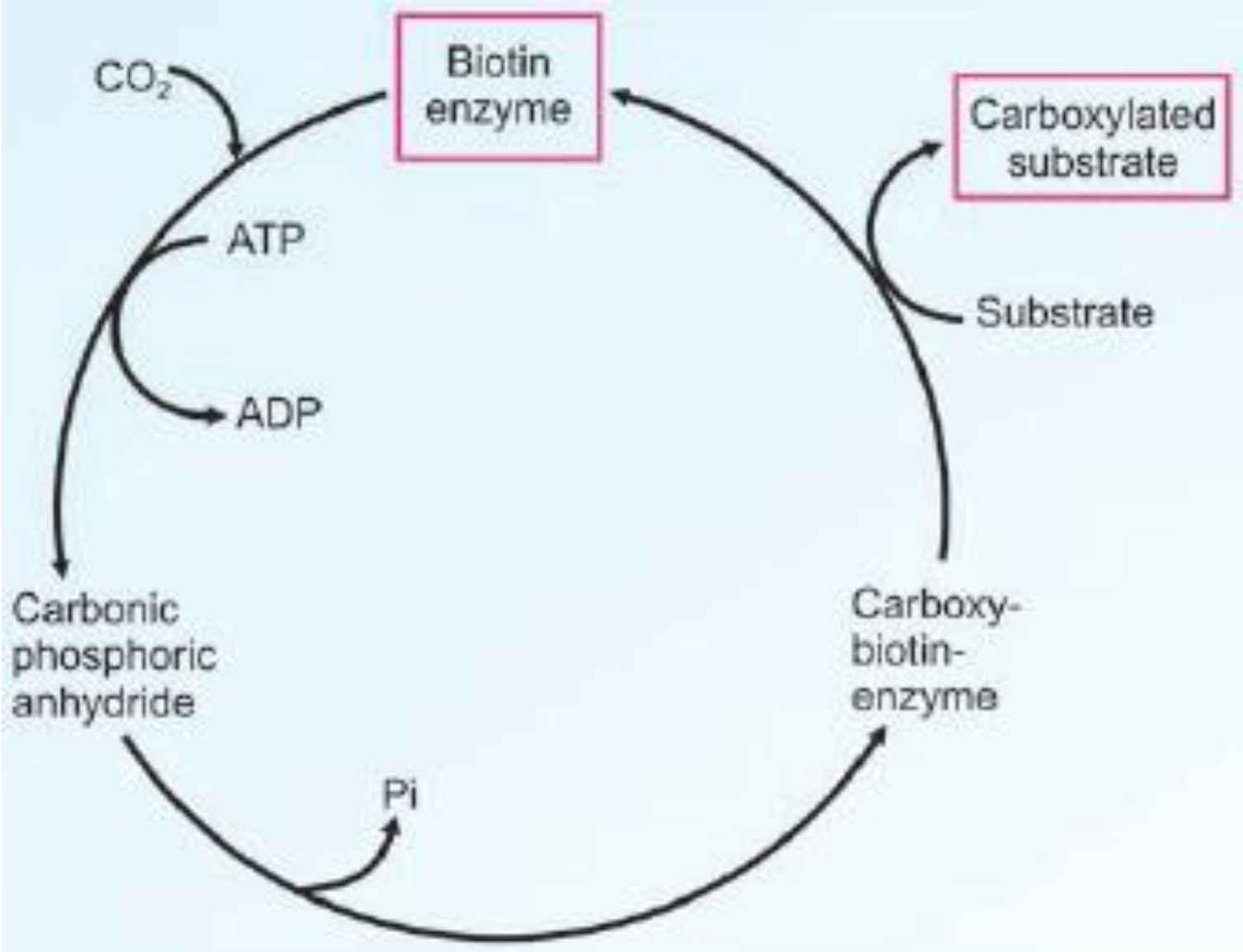
Vitamin B7/Biotin

Co-enzyme Activity of Biotin

- Biotin acts as co-enzyme for **carboxylation reactions**
- Biotin captures a molecule of CO2 which is **attached to nitrogen of the biotin molecule**
- The **energy required** for this reaction is provided by ATP
- Then the **activated carboxyl group is transferred** to the substrate.

Vitamin B7/Biotin

Co-enzyme Activity of Biotin



Vitamin B7/Biotin

Biotin Requiring CO2 Fixation Reactions

Acetyl CoA carboxylase

- This enzyme adds CO2 to acetyl CoA to form malonyl CoA
- This is the rate limiting reaction in biosynthesis of fatty acids

Acetyl CoA + CO2 + ATP → Malonyl CoA + ADP + Pi

Vitamin B7/Biotin

Biotin Requiring CO2 Fixation Reactions

Propionyl CoA carboxylase

Propionyl CoA + CO2 + ATP → Methyl malonyl CoA + ADP + Pi

Pyruvate carboxylase

Pyruvate + CO2 + ATP → Oxaloacetate + ADP + Pi

- This is important in two aspects. One, it provides the oxaloacetate, which is the catalyst for TCA cycle
- Second, it is an important enzyme in the gluconeogenic pathway

Vitamin B7/Biotin

Deficiency of Biotin

- Prolonged use of antibacterial drugs
- Biotin deficiency symptoms include dermatitis, atrophic glossitis, hyperesthesia, muscle pain, anorexia and hallucinations
- Injection of biotin 100-300 mg will bring about rapid cure of these symptoms.

Vitamin B7/Biotin

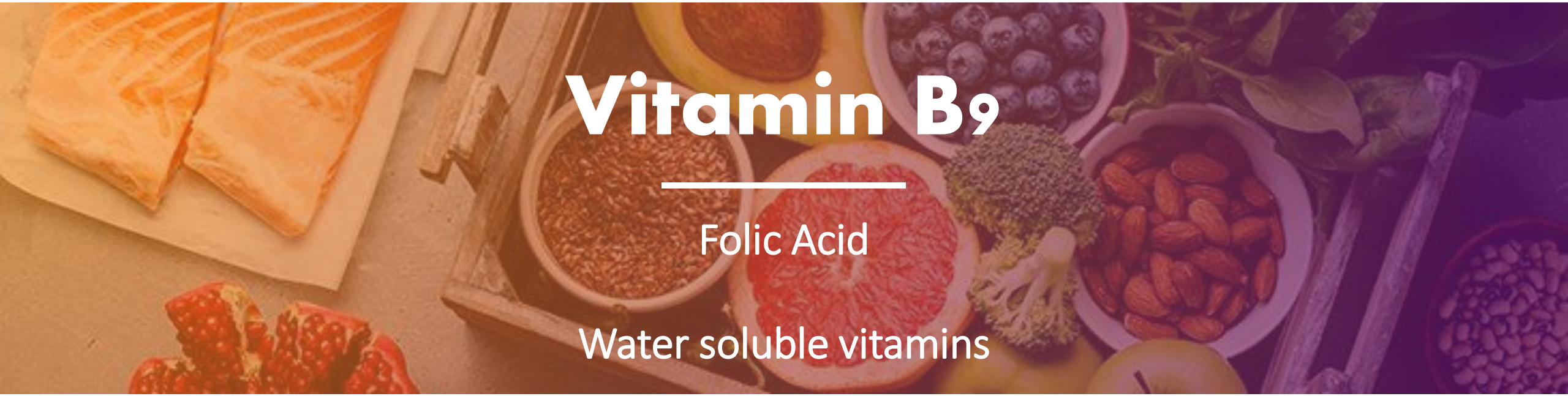


Requirement of Biotin

- About 200-300 mg will meet the daily requirements.

Sources of Biotin

- Normal bacterial flora of the gut will provide adequate quantities of biotin
- Moreover, it is distributed ubiquitously in plant and animal tissues
- Liver, yeast, peanut, soybean,
- milk and egg yolk are rich sources



Vitamin B_9

Folic Acid

Water soluble vitamins

Vitamin B9/Folic Acid

- The Latin word folium means leaf of vegetable
- Folic acid is abundant in vegetables. It is composed of three constituents

Absorption of Folic Acid

- Folic acid is readily absorbed by the upper part of jejunum
- In the blood, it is transported by beta globulins
- It is taken up by the liver where the coenzymes are produced
- Folic acid is not stored in tissues

Vitamin B9/Folic Acid

Co-enzyme Functions of Folic Acid

- The folic acid is first reduced to 7,8-dihydrofolic acid and further reduced to 5,6,7,8-tetrahydrofolic acid (THFA)
- Both reactions are catalyzed by NADPH dependent folate reductase
- The THFA is the carrier of one-carbon groups
- One carbon compound is an organic molecule that contains only a single carbon atom

Vitamin B9/Folic Acid

Co-enzyme Functions of Folic Acid

- The following groups are one carbon compounds:

i. Formyl (-CHO)

ii. Formimino (-CH=NH)

iii. Methenyl (-CH=)

iv. Methylene (-CH2–)

v. Hydroxymethyl (-CH2OH)

vi. Methyl (-CH3)

Vitamin B9/Folic Acid

Co-enzyme Functions of Folic Acid

- Methyl group in N5-methyl THFA is used for synthesis of active methionine, which takes part in transmethylation reactions
- Such transmethylation reactions are required for synthesis of choline, epinephrine, creatine

Vitamin B9/Folic Acid

Causes for Folate Deficiency

Pregnancy

- Folate deficiency is commonly seen in pregnancy, where requirement is increased

Defective absorption

- In sprue, celiac disease, gluten induced enteropathy, resection of jejunum and short-circuiting of jejunum in gastroileostomy, absorption is defective

Vitamin B9/Folic Acid

Causes for Folate Deficiency

Other reasons

- Drugs like anticovulsant drug
- Dietary deficiency
- Vit B12 deficiency

Vitamin B9/Folic Acid

Deficiency Manifestations

Reduced DNA synthesis

- In folate deficiency, THFA is reduced and thymidylate synthase enzyme is inhibited
- Hence dUMP is not converted to dTMP
- So dTTP is not available for DNA synthesis
- Thus cell division is arrested. Very rapidly dividing cells in bone marrow and intestinal mucosa are therefore most seriously affected

Vitamin B9/Folic Acid

Deficiency Manifestations

Macrocytic Anemia

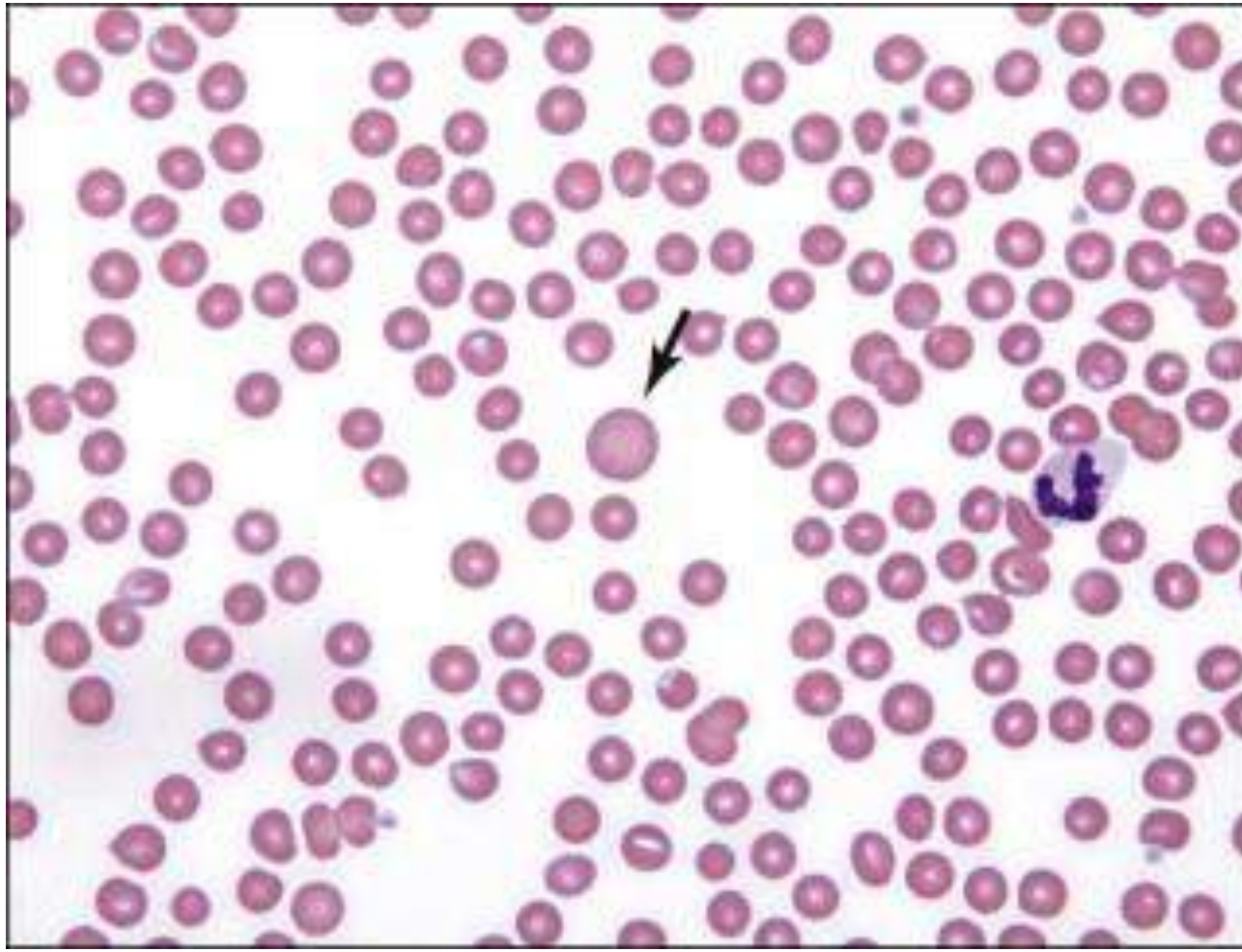
- It is the most characteristic feature of folate deficiency
- During erythropoiesis, DNA synthesis is delayed, but protein synthesis is continued
- Thus hemoglobin accumulates in RBC precursors
- This asynchrony or dissociation between the maturity of nucleus and cytoplasm is manifested as immature looking nucleus and mature eosinophilic cytoplasm in the bone marrow cells.

Vitamin B9/Folic Acid

Deficiency Manifestations

Macrocytic Anemia

- Reticulocytosis is a condition where there is an increase in reticulocytes, immature red blood cells
- These abnormal RBCs are rapidly destroyed in spleen
- This hemolysis leads to the reduction of lifespan of RBC
- Reduced generation and increased destruction of RBCs result in anemia.

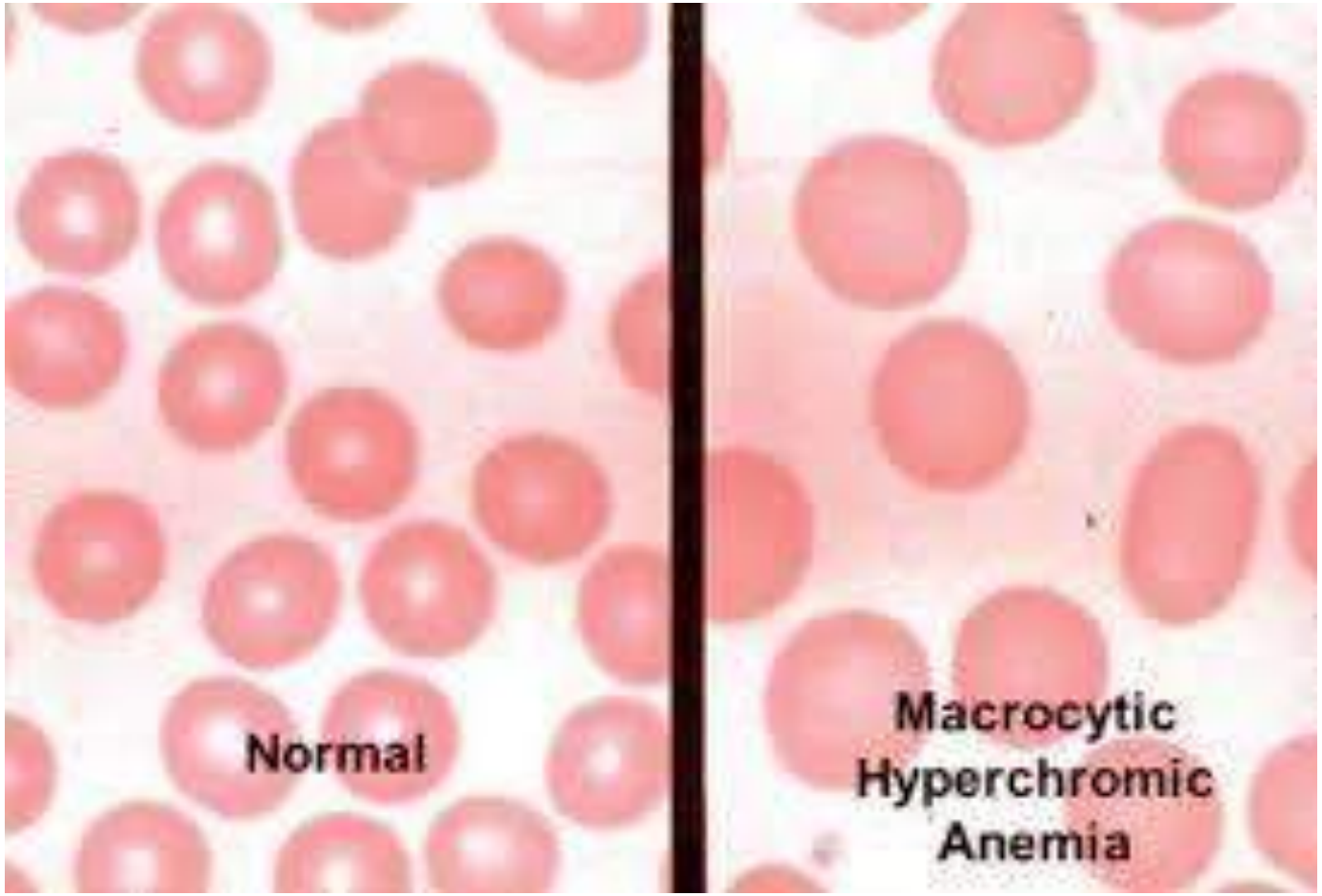


Vitamin B9/Folic Acid

Deficiency Manifestations

Macrocytic Anemia

- Leukopenia and thrombocytopenia are also manifested



Vitamin B9/Folic Acid

Deficiency Manifestations

Homocysteinemia

- Folic acid deficiency may cause increased homocysteine levels in blood
- Plasma homocysteine levels above 15 micromoles / L is known to increase the risk of coronary artery diseases

Vitamin B9/Folic Acid

Deficiency Manifestations

Birth Defects

- Folic acid deficiency during pregnancy may lead to homocysteinemia and neural tube defects in the fetus
- Folic acid prevents birth defects (fetal malformations such as spina bifida)
- So, supplementation of folic acid from early pregnancy is a must to prevent neural tube defects in the child

Vitamin B9/Folic Acid

Sources of Folic Acid

- Rich sources of folate are yeast, green leafy vegetables
- Moderate sources are cereals, pulses, oil seeds and egg
- Milk is a poor source for folic acid.



Vitamin B9/Folic Acid

Recommended Daily Allowance (RDA)

- The requirement of free folate is 200 microgram/day
- In pregnancy the requirement is increased to 400 microgram/day and during lactation to 300 microgram/day.



Vitamin B_{12}

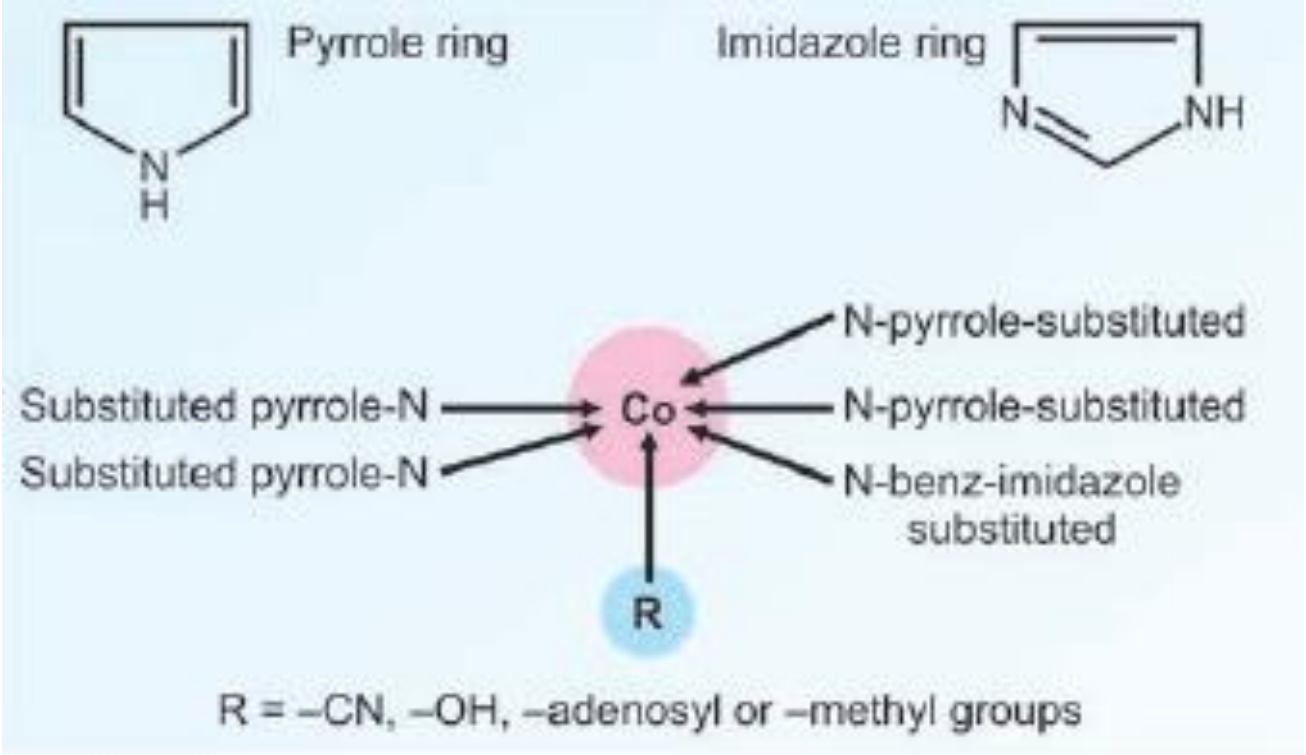
Cobalamins

Water soluble vitamins

Vitamin B12/Cobalamins

- Synonyms are cobalamin, extrinsic factor (EF) of Castle and antipernicious anemia factor

- **Cyanocobalamin:** When cyanide is added at the R position
- **Hydroxy cobalamin:** When hydroxyl group is attached at the R position
- **Adenosyl cobalamin (Ado-B12):** When taken up by the cells, these groups are removed and deoxy adenosyl cobalamin or Ado- B12 is formed
- **Methyl cobalamin:** When the methyl group replaces adenosyl group, it is known as methyl cobalamin



Vitamin B12/Cobalamins

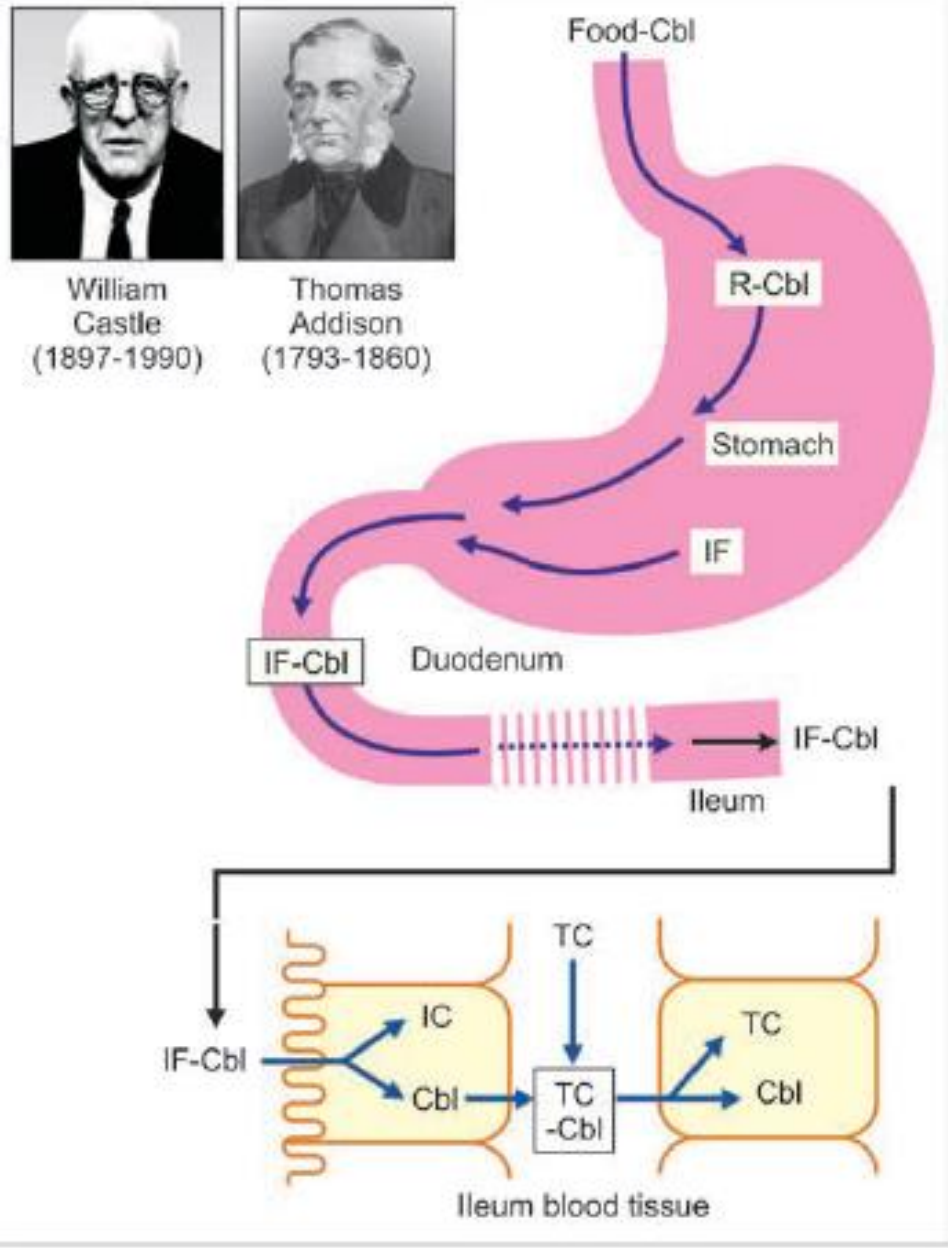
Absorption of Vitamin B12

- Absorption of vitamin B12 requires two binding proteins. First is the intrinsic factor (IF) of Castle
- The B12 is otherwise known as extrinsic factor (EF), that is, the factor derived from external sources
- Intrinsic factor is secreted by the gastric parietal cells. It is a glycoprotein with a molecular weight of 50,000

Vitamin B12/Cobalamins

Absorption of Vitamin B12

- One molecule of IF can combine with **2 molecules of B12**. This IF-B12 complex is attached with specific receptors on mucosal cells
- The whole IF-B12 complex is internalized
- It may be noted that B12 is absorbed from ileum, while folic acid is from jejunum



Vitamin B12/Cobalamins

Causes of B12 Deficiency

Nutritional

- Nutritional vitamin B12 deficiency is very common in India, especially among vegetarians of low socioeconomic group
- The only source for B12 in vegetarian diet is curd / milk, and lower income group may not be able to afford it

Vitamin B12/Cobalamins

Causes of B12 Deficiency

Decrease in absorption

- Absorptive surface is reduced by gastrectomy, resection of ileum and malabsorption syndromes

Vitamin B12/Cobalamins

Causes of B12 Deficiency

Addisonian pernicious anemia

- It is very rare in India, but common in European countries
- It is manifested usually in persons over 40 years
- It is an autoimmune disease with a strong familial background
- Antibodies are generated against IF
- So, IF becomes deficient, leading to defective absorption of B12

Vitamin B12/Cobalamins

Causes of B12 Deficiency

Gastric atrophy

- Although true Addisonian pernicious anemia is rare, similar atrophy of gastric epithelium leading to deficiency of IF and decreased B12 absorption is common in India
- In chronic iron deficiency anemia, there is generalized mucosal atrophy
- In about 40% cases of iron deficiency anemia, superadded gastric atrophy is seen.

Vitamin B12/Cobalamins

Causes of B12 Deficiency

Pregnancy

- Increased requirement of vitamin in pregnancy is another common cause for vitamin B12 deficiency in India

Fish tapeworm

- Although not seen in India, the fish tapeworm, *Diphillobothrium latum* infection is common in Scandinavian countries where eating live fish is a delicacy
- This tapeworm has a special affinity to B12 causing reduction in available vitamin

Vitamin B12/Cobalamins

Causes of B12 Deficiency

Megaloblastic anemia

- Megaloblastic anemia is a condition in which the bone marrow produces unusually large, structurally abnormal, immature red blood cells (megaloblasts)
- Bone marrow, the soft spongy material found inside certain bones, produces the main blood cells of the body -red cells, white cells, and platelets.
- Megaloblastic anemia has several different causes - deficiencies of either cobalamin (vitamin B12) or folate (vitamin B9) are the two most common causes. These vitamins play an essential role in the production of red blood cells.

Vitamin B12/Cobalamins

Causes of B12 Deficiency

Megaloblastic anemia

- The most common causes of megaloblastic anemia are deficiency of either cobalamin (vitamin B12) or folate (vitamin B9)
- These two vitamins serve as building blocks and are essential for the production of healthy cells such as the precursors to red blood cells
- Without these essential vitamins, the creation (synthesis) of deoxyribonucleic acid (DNA), the genetic material found in all cells, is hampered.

Vitamin B12/Cobalamins

Causes of B12 Deficiency

Signs & Symptoms

- Symptoms common to anemia usually develop at some point and may include fatigue, paleness of the skin (pallor), shortness of breath, lightheadedness, dizziness and a fast or irregular heartbeat
- The specific symptoms present in each individual can vary greatly.
- common symptoms include aches and pains, muscle weakness, and difficulty breathing (dyspnea). Individuals with megaloblastic anemia may also develop gastrointestinal abnormalities including diarrhea, nausea, and loss of appetite.

Vitamin B12/Cobalamins

Causes of B12 Deficiency

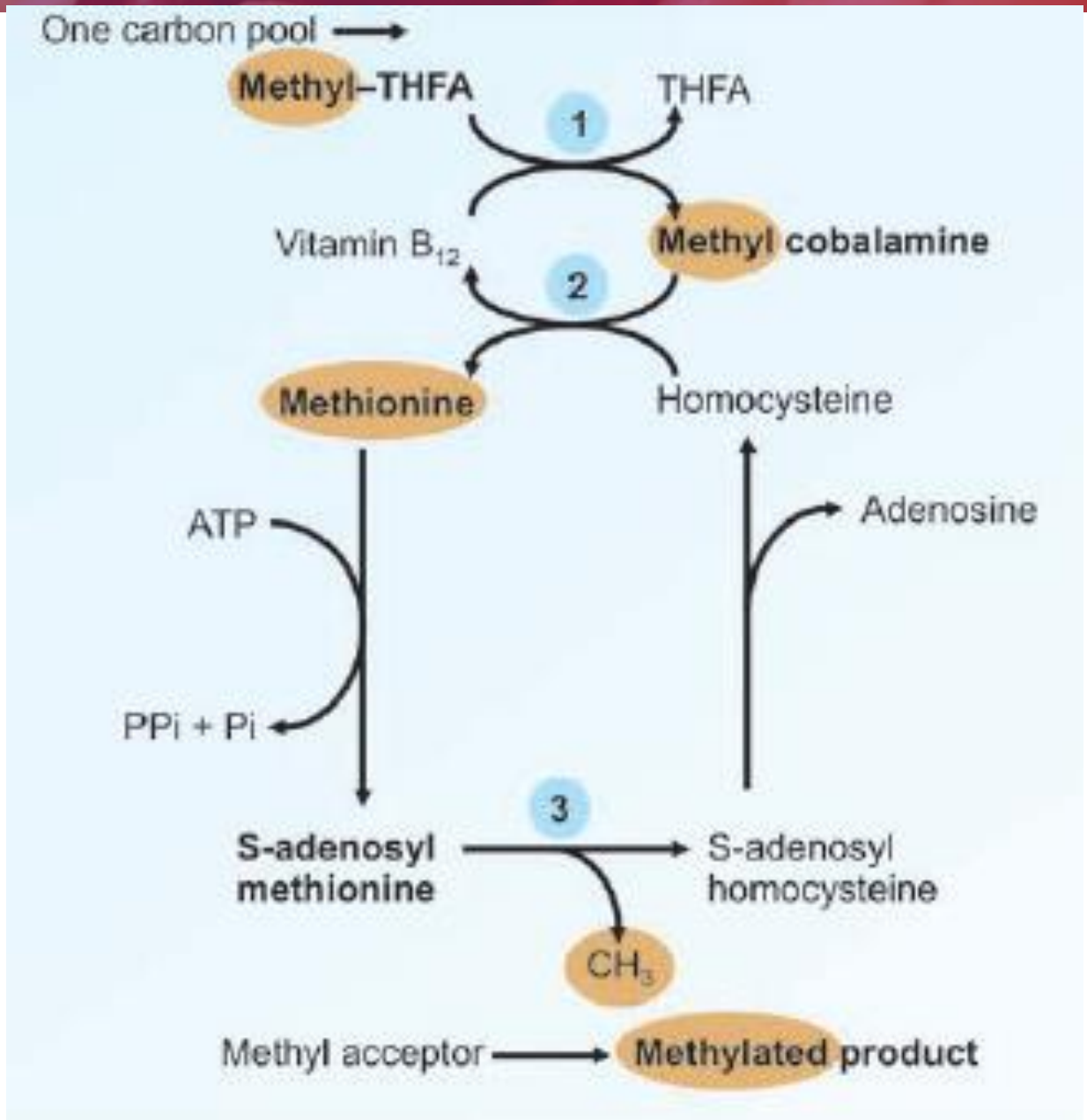
Signs & Symptoms

- Megaloblastic anemia resulting from cobalamin deficiency may also be associated with neurological symptoms. The initial neurological symptom may be tingling or numbness in the hands or feet. Additional symptoms develop over time including balance or gait problems, vision loss due to degeneration (atrophy) of the nerve that transmits impulses from the retina to the brain (optic nerve), and mental confusion or memory loss

Vitamin B12/Cobalamins

Demyelination

- In vitamin B12 deficiency, step 3 is also suppressed due to the non-availability of active methionine
- Therefore methylation of phosphatidyl ethanolamine to phosphatidyl choline is not adequate
- This leads to deficient formation of myelin sheaths of nerves, demyelination and neurological lesions





Vitamin C

Ascorbic Acid

Water soluble vitamins

Vitamin C/Ascorbic acid

- It is water soluble and is easily destroyed by heat, alkali and storage. In the process of cooking, 70% of vitamin C is lost
- Ascorbic acid is readily absorbed from gastrointestinal tract. The vitamin is excreted in urine

Biochemical Functions of Vitamin C

Reversible oxidation-reduction

- It can change between ascorbic acid and dehydroascorbic acid
- Most of the physiological properties of the vitamin could be explained by this redox system.

Vitamin C/Ascorbic acid

Biochemical Functions of Vitamin C

Hydroxylation of proline and lysine

- Ascorbic acid is necessary for the post-translational hydroxylation of proline and lysine residues
- Hydroxyproline and hydroxylysine are essential for the formation of cross links in the collagen, which gives the tensile strength to the fibres
- This process is absolutely necessary for the normal production of supporting tissues such as osteoid, collagen and intercellular cement substance of capillaries

Vitamin C/Ascorbic acid

Biochemical Functions of Vitamin C

Tryptophan metabolism

- Ascorbic acid is necessary for the hydroxylation of tryptophan to 5-hydroxy tryptophan
- This is required for the formation of serotonin

Tyrosine metabolism

- Vitamin C helps in the oxidation of parahydroxy phenyl pyruvate to homogentisic acid

Vitamin C/Ascorbic acid

Biochemical Functions of Vitamin C

Iron metabolism

- Ascorbic acid enhances the iron absorption from the intestine
- Ascorbic acid reduces ferric iron to ferrous state, which is preferentially absorbed

Hemoglobin metabolism

- It is useful for re-conversion of met-hemoglobin to hemoglobin

Vitamin C/Ascorbic acid

Biochemical Functions of Vitamin C

Folic acid metabolism

- Ascorbic acid is helping the enzyme folate reductase to reduce folic acid to tetrahydrofolic acid
- Thus it helps in the maturation of RBC

Steroid synthesis

- Large quantities of vitamin C are present in adrenal cortex
- The ascorbic acid is depleted by ACTH stimulation
- So the vitamin has some role in adrenal steroidogenesis
- Vitamin C helps in the synthesis of bile acids from cholesterol

Vitamin C/Ascorbic acid

Biochemical Functions of Vitamin C

Phagocytosis

- Ascorbic acid stimulates phagocytic action of leukocytes and helps in the formation of antibodies

Anti-oxidant property

- As an anti-oxidant, it may prevent cancer formation
- Aniline dyes are known to induce bladder cancer in factory workers
- Daily intake of vitamin C reduces this risk for cancer

Vitamin C/Ascorbic acid

Biochemical Functions of Vitamin C

Cataract

- Vitamin C is concentrated in the lens of eye
- Regular intake of ascorbic acid reduces the risk of cataract formation

Vitamin C/Ascorbic acid

Deficiency Manifestations of Vitamin C

Scurvy

- Anemia, myalgia, or pain, including bone pain
- swelling, or edema
- petechiae, or small red spots resulting from bleeding under the skin
- gum disease and loss of teeth
- poor wound healing, shortness of breath
- mood changes, and depression

Gingival Hemorrhage



Corkscrew Hair



Periungual Hemorrhage



Vitamin C/Ascorbic acid

Deficiency Manifestations of Vitamin C

Infantile scurvy (Barlow's disease)

- In infants between 6 to 12 months of age, (period in which weaning from breast milk), the diet should be supplemented with vitamin C sources
- Otherwise, deficiency of vitamin C is seen

Vitamin C/Ascorbic acid

Deficiency Manifestations of Vitamin C

Hemorrhagic tendency

- In ascorbic acid deficiency, collagen is abnormal and the intercellular cement substance is brittle
- So capillaries are fragile, leading to the tendency to bleed even under minor pressure
- Subcutaneous hemorrhage may be manifested as petechiae in mild deficiency and as ecchymoses or even hematoma in severe conditions
- If a sphygmomanometer cuff is placed in the forearm, and the pressure is kept for 5 minutes, petechiae may be seen under the skin. This is a useful clinical test.

Vitamin C/Ascorbic acid

Deficiency Manifestations of Vitamin C

Internal hemorrhage

- In severe cases, hemorrhage may occur in the conjunctiva and retina
- Internal bleeding may be seen as epistaxis, hematuria or melena
- **Epistaxis** also called acute hemorrhage or nose bleed is a medical condition in which bleeding occurs from the nasal cavity of the nostril
- **Hematuria** is the presence of red blood cells in the urine
- **Melena** refers to the dark black, tarry feces that are associated with upper gastrointestinal bleeding

Vitamin C/Ascorbic acid

Deficiency Manifestations of Vitamin C

Oral cavity

- In severe cases of scurvy, the gum becomes painful, swollen and spongy
- The pulp is separated from the dentine and finally teeth are lost
- Wound healing may be delayed

Vitamin C/Ascorbic acid

Deficiency Manifestations of Vitamin C

Bones

- In the bones, the deficiency results in the failure of the osteoblasts to form the intercellular substance, Osteoid
- Without the normal ground substance, the deposition of bone is arrested
- The resulting scorbutic bone is weak and fractures easily
- There may be hemorrhage into joint cavities
- Painful swelling of joints may prevent locomotion of the patient

Vitamin C/Ascorbic acid

Deficiency Manifestations of Vitamin C

Anemia

- Loss of blood by hemorrhage
- Decreased iron absorption
- Decreased tetrahydrofolic acid
- Accumulation of met-hemoglobin.

Vitamin C/Ascorbic acid

Deficiency Manifestations of Vitamin C

Dietary Sources of Vitamin C

- Rich sources are amla (Indian gooseberry) (700 mg/ 100 g), guava (300 mg/100 g), lime, lemon and green leafy vegetables



Requirement of Vitamin C

- Recommended daily allowance is 75 mg/day (equal to 50 ml orange juice)
- During pregnancy, lactation, and in aged people requirement may be 100 mg/ day.

Fat soluble Vitamins

Biological function and Disease manifestation of water and fat soluble vitamins (no structures)

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Vitamins

The vitamins are mainly classified into two:

1. The fat soluble vitamins are A, D, E and K
2. Water soluble vitamins are named as B complex and C

	Fat Soluble Vitamins	Water Soluble Vitamins
Solubility in fat	Soluble	Not Soluble
Water Solubility	Not soluble	Soluble
Absorption	Along with lipids Requires Bile salts	Absorption simple
Carrier proteins	Present	No carrier proteins
Storage	Stored in liver	No storage
Excretion	Not excreted	Excreted
Deficiency	Manifests only when stores are depleted	Manifests rapidly as there is no storage
Toxicity	Hypervitaminosis may result	Unlikely, since excess is excreted
Treatment of deficiency	Single large doses may prevent deficiency	Regular dietary supply is required
Major Vitamins	A,D,E and K	B and C

Vitamin deficiency

In general, deficiency of vitamins may occur due to:

a. reduced intake

b. impaired absorption

c. impaired metabolism

d. additional requirements

e. increased losses.



Vitamin A

Fat soluble vitamins

Vitamin A

- Vitamin A is **fat soluble**.
- The active form is present only in animal tissues.
- The pro-vitamin, beta-carotene is present in plant tissues. Beta carotene has two beta ionone rings connected by a polyprenoid chain
- All the compounds with vitamin A activity are referred to as retinoids. They are polyisoprenoid compounds having a beta-ionone ring system
- Three different compounds with vitamin A activity are **retinol** (vitamin A alcohol), **retinal** (vitamin A aldehyde) and **retinoic acid** (vitamin A acid)

Vitamin A

- The retinal may be reduced to retinol by **retinal reductase**. This reaction is **readily reversible**. **Retinal is oxidized to retinoic acid**, which cannot be converted back to the other forms
- The side chain contains **alternate double bonds**, and hence many isomers are possible. The **all-trans variety of retinal, also called vitamin A1** is most common
- **Vitamin A2** is found in fish oils and has an extra double bond in the ring. Biologically important compound is **11-cis-retinal**.

Vitamin A

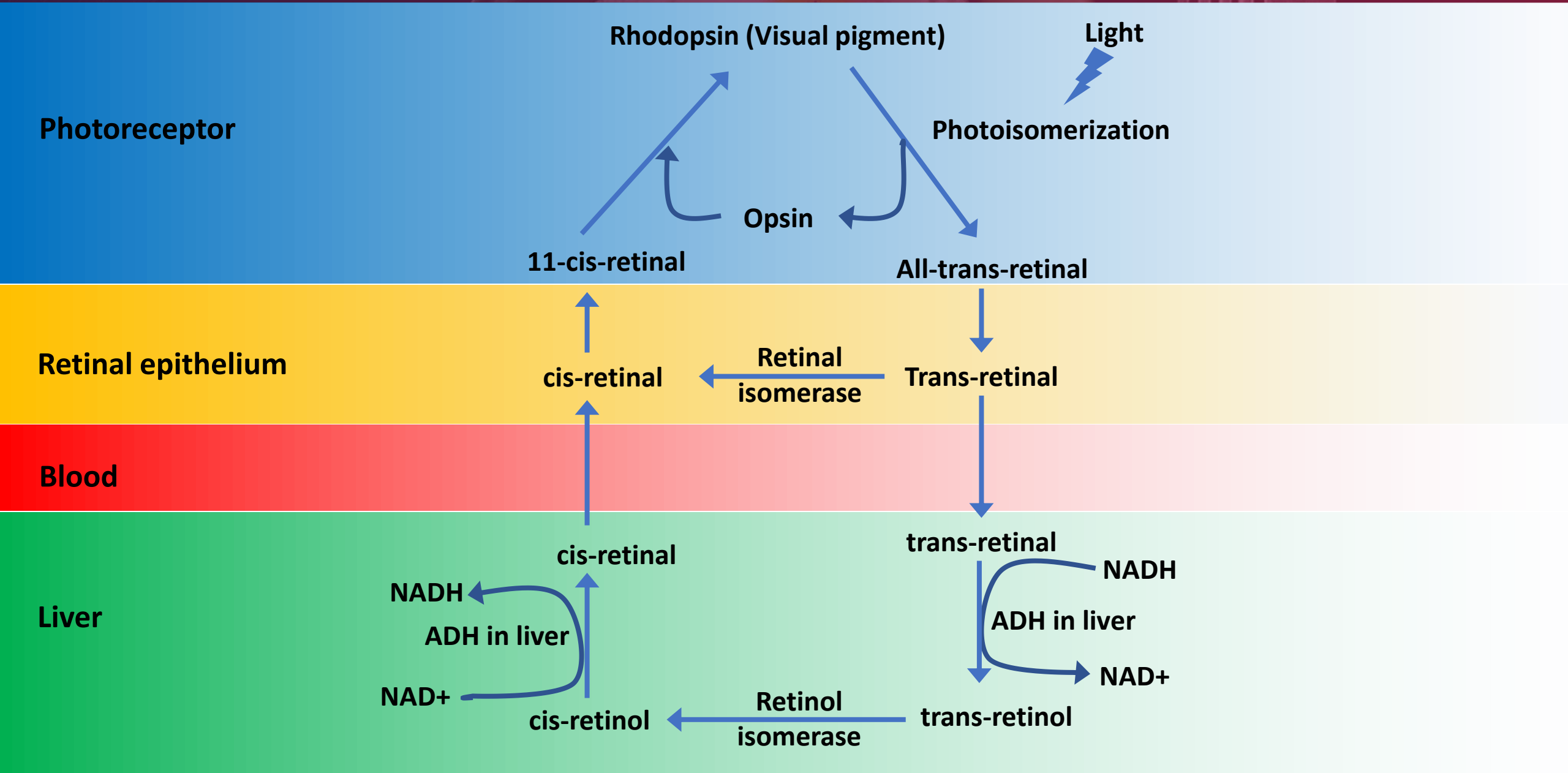
Absorption of Vitamin A

I. Beta carotene is cleaved by a **di-oxygenase**, to form **retinal.** The **retinal is reduced to retinol** by an NADH or NADPH dependent **retinal reductase** present in the intestinal mucosa. Intestine is the major site of absorption

II. The absorption is along with other fats and **requires bile salts**. In biliary tract obstruction and steatorrhoea, vitamin A absorption is reduced.

III. Within the mucosal cell, the **retinol is reesterified with fatty acids**, incorporated into **chylomicrons** and **transported to liver**. In the liver stellate cells, **vitamin is stored as retinol palmitate**.

Vitamin A


Photoreceptor
Rhodopsin (Visual pigment)
Light
Photoisomerization
Opsin
11-cis-retinal
All-trans-retinal
Retinal epithelium
cis-retinal
Retinal isomerase
Trans-retinal
Blood
Liver
cis-retinal
trans-retinal
NADH
ADH in liver
NAD+
NADH
ADH in liver
NAD+
cis-retinol
Retinol isomerase
trans-retinol

Vitamin A

Other Biochemical Functions of Vitamin A

- Retinoic acid has a role in the **regulation of gene expression** and **differentiation of tissues**
- **All-trans-retinoic acid** and **9-cisretinoic acid** act like steroid hormones
- They bind to nuclear receptors; retinoic acid along with the receptor binds to the response elements of DNA. **Retinoic acid receptors (RAR)** bind all-trans-retinoic acid, while **retinoic x receptors (RXR)** bind to 9-cisretinoic acid
- **RXRs** form dimers with **vitamin D-receptor** also. This explains why **deficiency of vitamin A impairs vitamin D function**

Vitamin A

Reproductive system

- Retinol is **necessary for the reproductive system**
- Retinol **acts like a steroid hormone** in controlling the expression of certain genes
- This may account for the requirement of **vitamin A for normal reproduction**
- In vitamin deficiency, **miscarriages** are noticed in female rats

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- They function as **antioxidant property** and functions as a **anticancer agent**
- Beta carotenes may be **useful in preventing heart attacks**
- Vitamin A is necessary for the **maintenance of normal epithelium and skin.**

Vitamin A

Deficiency manifestations of Vitamin A

Night Blindness or Nyctalopia

Visual acuity is diminished in dim light. The patient cannot read or drive a car in poor light. The **dark adaptation time is increased.**

Xerophthalmia

- The **conjunctiva becomes dry, thick and wrinkled**. The conjunctiva gets **keratinized and loses its normal transparency**. Dryness spreads to cornea. It becomes **glazy and lusterless due to keratinization** of corneal epithelium. **Infections may follow**

Vitamin A

Deficiency manifestations of Vitamin A

Bitot's Spots

- These are seen as **greyish-white triangular plaques** firmly adherent to the conjunctiva. This is due to **increased thickness of conjunctiva** in certain areas. All the ocular changes mentioned so far are **completely reversible** when vitamin is supplemented

Keratomalacia

- When the xerophthalmia persists for a long time, it progresses to **keratomalacia (softening of the cornea).** There is **degeneration of corneal epithelium** which may get vascularized. Later, **corneal opacities develop**. Bacterial **infection leads to corneal ulceration**, perforation of cornea and **total blindness**.

Vitamin A

Deficiency manifestations of Vitamin A

Preventable Blindness

- The deficiency of vitamin A is the **most common cause of blindness in Indian children** below the age of 5. One-third of the world's blind population are residing in India. **About 40% of blindness is preventable.**

  Vitamin A deficiency is a major public health problem. A single dose of vitamin A is given, as a prophylactic measure, to children below 1 year age.

Vitamin A

Deficiency manifestations of Vitamin A

Skin and Mucous Membrane Lesions

- The **skin becomes rough**. Keratinizing of the **epithelium of the respiratory, gastrointestinal and genitourinary tract** have been observed. **Epithelium is atrophied**. Keratinization of urinary tract epithelium may lead to **kidney stone**
- The alterations in skin may cause increased occurrence of **generalized infections**. Therefore in old literature, vitamin A is referred to as **anti-inflammatory vitamin**
- Isoretinone, a synthetic variant of vitamin A is known to reduce the sebaceous secretions, hence it is used to **prevent acne formation during adolescence**.

Vitamin A

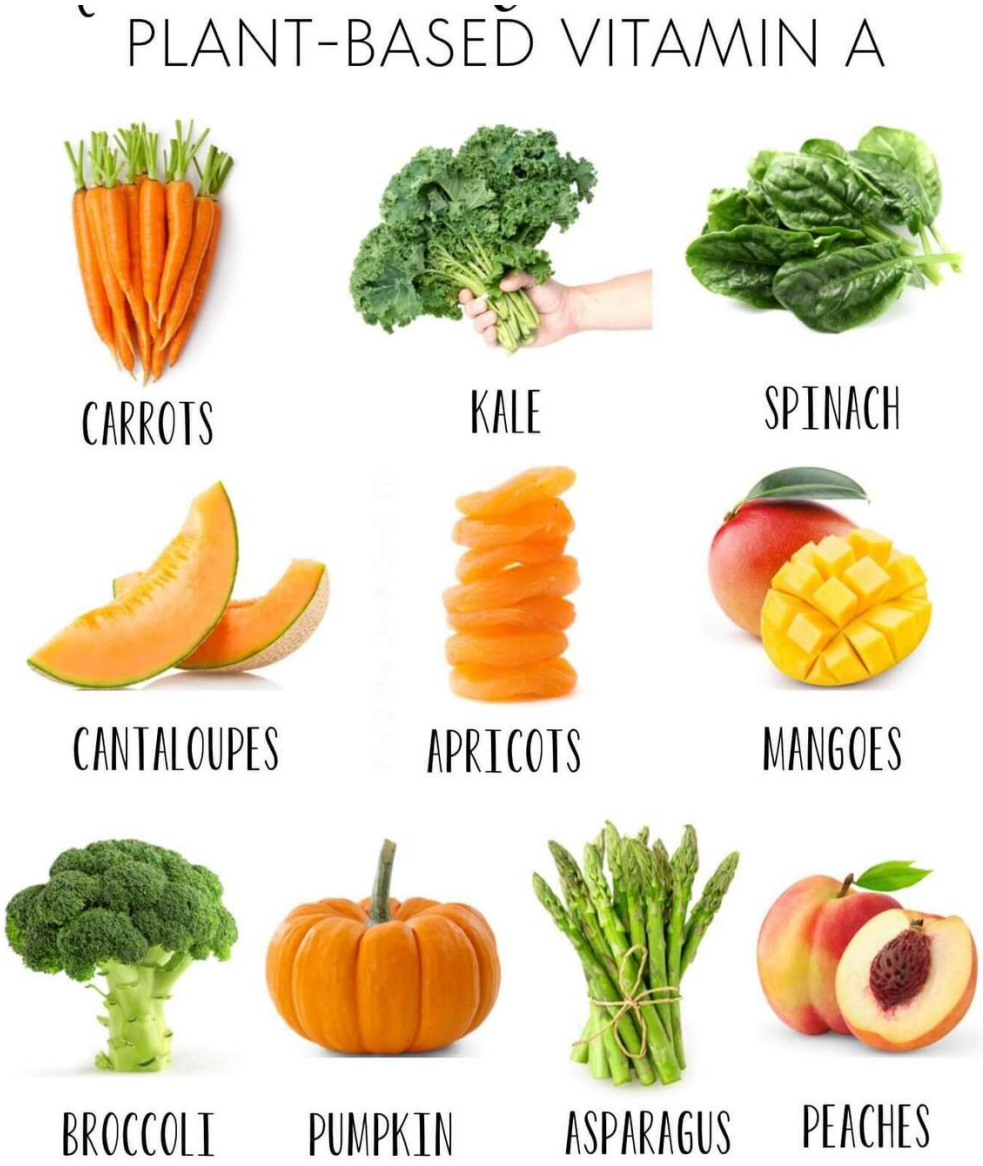
Causes for Vitamin A Deficiency

1. Decreased intake
2. Obstructive jaundice causing defective absorption
3. Cirrhosis of liver leading to reduced synthesis of RBP
4. Severe malnutrition, where amino acids are not available for RBP synthesis
5. Chronic nephrosis, where RBP is excreted through urine.

Vitamin A

Dietary Sources of Vitamin A

- Animal sources include milk, butter, cream, cheese, egg yolk and liver. Fish liver oils (cod liver oil and shark liver oil) are very rich sources of the vitamin.
- Vegetable sources contain the yellow pigment beta carotene.
- Carrot contains significant quantity of beta carotene . Papaya, mango, pumpkins and green leafy vegetables (spinach, amaranth) are other good sources for vitamin A activity. During cooking the activity is not destroyed.



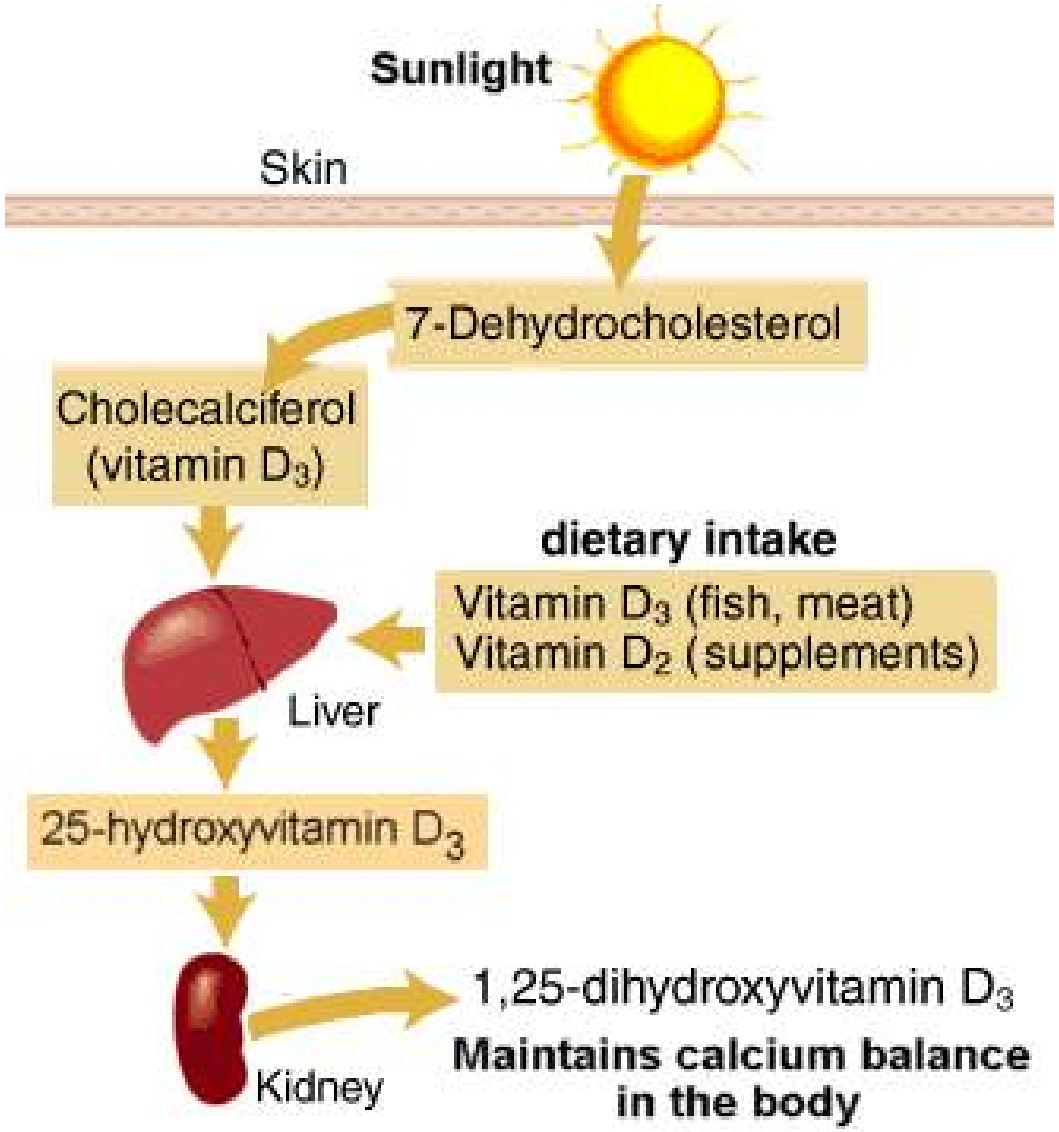


Vitamin D

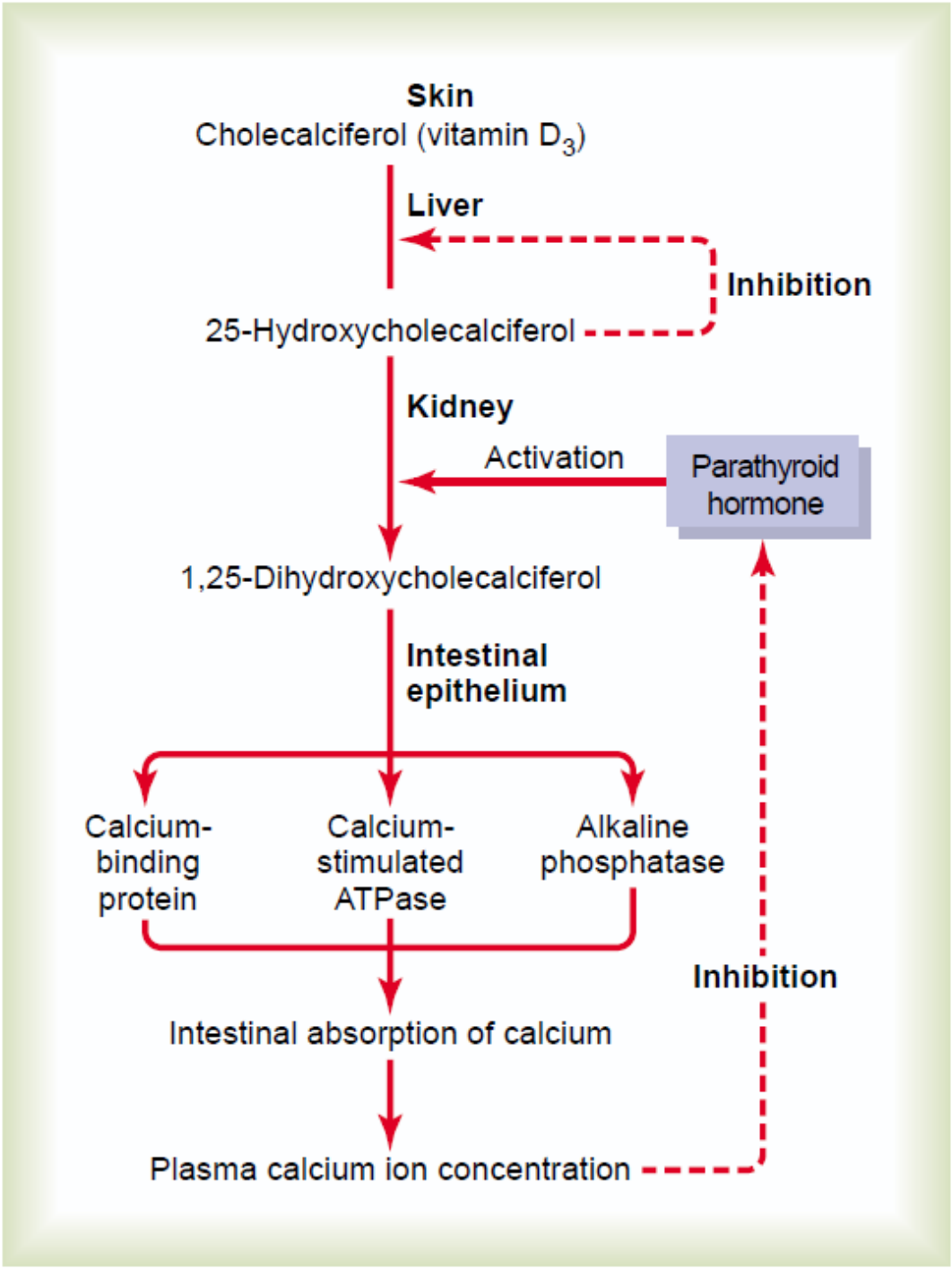
Fat soluble vitamins

Vitamin D or cholecalciferol

Formation of Vitamin D



Formation of Vitamin D



Vitamin D

Activation of Vitamin D

1. Vitamin D is a **prohormone**. The cholecalciferol is **first transported to liver**, where hydroxylation at 25th position occurs, to form **25-hydroxy cholecalciferol** (25-HCC). The hepatic 25-hydroxylase is a microsomal monooxygenase. It requires cytochrome P-450 and NADPH. **25-HCC is the major storage form.**

2. In plasma, 25-HCC is bound to "**vitamin D binding protein**" (VDBP), an alpha-2 globulin.

3. In the kidney, it is further hydroxylated at the 1st position. The 1-alpha hydroxylase is located in mitochondria of proximal convoluted tubules. It requires cytochrome P-450, NADPH and ferrodoxin (an iron-sulfur protein). Thus **1,25-dihydroxy cholecalciferol (DHCC) is generated**. Since it contains three hydroxyl groups at 1, 3 and 25 positions, it is also called **Calcitriol**. The calcitriol thus formed is the **active form of vitamin**; it is a hormone.

Vitamin D

Biochemical Effects of Vitamin D

The sites of action are:

1. **intestinal villi cells**
2. **bone osteoblasts**
3. **kidney distal tubular cells**

Vitamin D

Vitamin D and Absorption of Calcium

- Calcitriol promotes the absorption of calcium and phosphorus from the intestine
- In the brush-border surface, calcium is absorbed passively
- From the intestinal cell to blood, absorption of calcium needs energy. It is either by the sodium-calcium exchange mechanism or by pumping out the calcium-calbindin complex
- Calcitriol helps in transcription of specific genes that code for Calbindin. Due to the increased availability of calcium binding protein, the absorption of calcium is increased.

Vitamin D

Effect of Vitamin D in Bone

- Mineralization of the bone is increased by increasing the activity of osteoblasts
- Calcitriol coordinates the remodelling action of osteoclasts and osteoblasts. It produces the differentiation of osteoclast precursors from multinucleated cells of osteoblast lineage
- Calcitriol stimulates osteoblasts which secrete alkaline phosphatase
- Due to this enzyme, the local concentration of phosphate is increased
- The ionic product of calcium and phosphorus increases, leading to mineralization

Vitamin D

Effect of Vitamin D in Renal Tubules

- Calcitriol increases the reabsorption of calcium and phosphorus by renal tubules, therefore both minerals are conserved

Deficiency of Vitamin D

- The deficiency diseases are **rickets in children** and **osteoporosis in adults**. Hence vitamin D is known as **antirachitic vitamin**



Vitamin D

Causes for Vitamin D Deficiency

- Deficiency of vitamin D can occur in people who are **not exposed to sunlight properly**, e.g. inhabitants of northern latitudes, in winter months, in people who are bedridden for long periods, or those who cover the whole body
- **Nutritional deficiency of calcium or phosphate** may also produce similar clinical picture
- **Malabsorption of vitamin** (obstructive jaundice and steatorrhea). High **phytate** content in diet may also reduce the absorption of vitamin
- Abnormality of vitamin D activation. **Liver and renal diseases** may retard the hydroxylation reactions.
- **Deficient renal absorption of phosphates**

Vitamin D

Clinical Features of Rickets

- Rickets is seen in children. There is insufficient mineralization of bone. Bones become soft. The bone growth is markedly affected
- Plasma calcium and phosphorus are low normal with alkaline phosphatase (bone isoenzyme) being markedly elevated
- The classical features of rickets are bone deformities. Weight bearing bones are bent
- beading of ribs or rickety rosary, bossing of frontal bones, and pigeon chest

Vitamin D

Requirement of Vitamin D

1. Children = 10 microgram (400 IU)/day
2. Adults = 5 microgram (200 IU)/day
3. Pregnancy, lactation = 10 microgram/day
4. Above the age of 60 = 600 IU per day

Vitamin D

Sources of Vitamin D

- Exposure to sunlight produces cholecalciferol
- Moreover fish liver oil, fish and egg yolk are good sources of the vitamin
- Milk contains moderate quantity of the vitamin

Vitamin D

Hypervitaminosis D

- Doses above 1500 units per day for very long periods may cause toxicity
- Symptoms include weakness, polyuria, intense thirst, difficulty in speaking, hypertension and weight loss
- Hypercalcemia leads to calcification of soft tissues, (metastatic calcification, otherwise called calcinosis), especially in vascular and renal tissues
- Although vitamin D is toxic in higher doses, excessive exposure to sunlight does not result in vitamin D toxicity, because excess D3 is destroyed by sunlight itself



Vitamin E

Fat soluble vitamins

Vitamin E

Biochemical Role of Vitamin E

- Vitamin E is the **most powerful natural antioxidant**. Free radicals are continuously being generated in living systems. Their prompt inactivation is of great importance
- Vitamin E is a known **biological antioxidant** able to **quench the lipid peroxidation** chain and to **protect the plasma membranes from the attack of free radicals**

Vitamin E

Biochemical Role of Vitamin E

- The free radicals would attack bio-membranes. **Vitamin E protects RBC from hemolysis**. By preventing the peroxidation, it keeps the structural and functional integrity of all cells
- Gradual deterioration of ageing process is due to the cumulative effects of free radicals. Vitamin E also **boosts immune response**
- It **reduces the risk of atherosclerosis** by reducing oxidation of LDL. Vitamin E can **depress leukocyte oxidative bactericidal activity**

Vitamin E

Inter-relationship with Selenium

- Selenium is present in **glutathione peroxidase**; an important enzyme that oxidizes and destroys the free radicals
- Selenium has been **found to decrease the requirement of vitamin E and vice versa**. They act synergistically to minimize lipid peroxidation.

Vitamin E

Deficiency Manifestations of Vitamin E

- In rats, **inability to produce healthy ovum** and **loss of motility of spermatozoa**, **hemolysis of red cells**, **acute hepatic necrosis** and **muscular dystrophy** are observed
- In a normal adult, the body **vitamin E stores** can meet the requirement for several months.
- **Human deficiency has not been reported**. But in volunteers, vitamin E deficiency has been shown to produce **increased fragility of RBCs**, **muscular weakness and creatinuria**.

Vitamin E

Recommended Daily Allowance

- Males **10 mg per day**
- Females **8 mg/day**
- Pregnancy **10 mg/day**
- Lactation **12 mg/day**
- 15 mg of vitamin E is equivalent to 33 international units. The requirement increases with higher intake of PUFA. Pharmacological dose is 200-400 IU per day



Vitamin K

Fat soluble vitamins

Vitamin K

Absorption and Storage

- Absorption of vitamin K **occurs in the intestine** along with chylomicrons
- **Bile salts are required** for the normal absorption
- The vitamin K may be **derived from the diet or intestinal bacterial synthesis**
- It is **stored in the liver** and **transported in plasma along with beta lipoproteins**

Vitamin K

Biochemical Role of Vitamin K

- Vitamin K is **necessary for coagulation**. Factors dependent on vitamin K are Factor II (prothrombin); Factor VII (SPCA); Factor IX (Christmas factor); Factor X (Stuart Prower factor)
- All these factors are synthesized by the liver as inactive zymogens. They undergo post-translational modification; gamma carboxylation of glutamic acid residues. These are the binding sites for calcium ions. The **gamma carboxy glutamic acid (GCG) synthesis requires vitamin K as a co-factor**

Vitamin K

Biochemical Role of Vitamin K

- Vitamin K dependent **gamma carboxylation** is also necessary for the functional activity of **osteocalcin as well as structural proteins of kidney, lung and spleen**
- Osteocalcin is synthesized by osteoblasts and **seen only in bone**. It is a small protein (40-50 amino acids length) that binds tightly to hydroxy apatite crystals of bone
- This binding is dependent on the degree of gamma carboxylation. **Osteocalcin also contains hydroxyproline**, so it is **dependent on both vitamins K and C**

Vitamin K Dependent Carboxylase

- It is **microsomal enzyme**. It requires oxygen, CO2, NADPH and reduced vitamin K. In this process, the vitamin passes through a cycle

Vitamin K

Causes for Deficiency of Vitamin K

- In normal adults dietary **deficiency seldom occurs** since the intestinal bacterial synthesis is sufficient to meet the needs of the body
- However deficiency can **occur in conditions of malabsorption of lipids**
- This can result from **obstructive jaundice, chronic pancreatitis, sprue**, etc. **Prolonged antibiotic therapy and gastrointestinal infections with diarrhea** will destroy the bacterial flora and can also lead to vitamin K deficiency.

Vitamin K

Clinical Manifestations of Deficiency

- **Hemorrhagic disease of the newborn** is attributed to vitamin K deficiency
- The newborns, especially the premature infants have relative vitamin K deficiency
- This is due to lack of hepatic stores, limited oral intake (breast milk has very low levels, 15 mg/liter) and absence of intestinal bacterial flora
- It is often advised that pre-term infants be given **prophylactic doses of vitamin K**
- In children and adults, Vitamin K deficiency may be **manifested as bruising tendency, ecchymotic patches, mucous membrane hemorrhage, post-traumatic bleeding and internal bleeding**
- **Prolongation of prothrombin time and delayed clotting time** are characteristic of vitamin K deficiency

Vitamin K

Clinical Manifestations of Deficiency

- Measurement of **prothrombin time (PT)** is taken as an index of liver function
- When liver function is considerably lowered, prolongation of PT occurs due to deficient synthesis of the coagulation factors
- In such cases, administration of vitamin fails to restore PT to normal levels. Hence before undertaking any surgery on jaundiced patients, **PT before and after administration of vitamin K should be done**
- **Warfarin** and **dicoumarol** will competitively **inhibit the gamma carboxylation** system due to structural similarity with vitamin K. Hence they are **widely used as anticoagulants** for therapeutic purposes.

Vitamin K

Daily Requirement of Vitamin K

- Recommended daily allowance is **50-100 mg/day.** This is usually available in a normal diet

Sources of Vitamin K

- **Green leafy vegetables** are good dietary sources
- Even if the diet does not contain the vitamin, intestinal bacterial synthesis will meet the daily requirements, as long as absorption is normal